Priyanshu Ranjan

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Professional Summary

Computer Science undergraduate specializing in AI & Machine Learning with a CGPA of **8.76/10**, experienced in building full-stack applications and backend services using Java, Spring Boot, Node.js, REST APIs, and SQL/NoSQL databases. Built role-based workflows, authentication, persistence, and automated tests, with a focus on backend development, API design, and reliable application behavior.

Education

Vellore Institute of Technology **Sep 2023 – Expected 2027**

B.Tech in Computer Science & Engineering (AI & ML Specialization) | ***CGPA: 8.76/10***

Projects

Swayam Saarthi - Ride-Hailing Platform **Dec 2025 – Jul 2026**

Full-Stack Web Application — React, Node.js, Express, MongoDB, JWT, Socket.IO

- Built a reverse ride-hailing platform with **24 React page components, 35+ REST API routes, 7 Mongoose models, and 3 role-based portals** for users, drivers, and administrators.
- Implemented a **5-state booking workflow** with JWT authentication and role-based access control across user, driver, and admin portals, connecting booking operations with ride updates, notifications, reviews, and a payment workflow.
- Integrated **Socket.IO** for real-time communication and Leaflet/OpenStreetMap for ride tracking; deployed the frontend, backend, and database using Vercel, Render, and MongoDB Atlas.

Library Management System **Jul – Aug 2026**

Java Backend Application — Java 21, Spring Boot, Spring Security, JPA, MySQL, JUnit 5

- Designed a layered Spring Boot backend with REST controllers, service-layer business logic, Spring Data JPA/Hibernate repositories, and MySQL persistence.
- Implemented session-based authentication with Spring Security and BCrypt, role-based authorization for administrator/librarian roles, and CRUD for 3 entities: books, students, and librarians.
- Built borrow/return workflows with availability checks and preserved transaction history; added server-side validation and MockMvc/repository tests covering authentication, CSRF, CRUD, and borrow/return scenarios.

VoiceCart - Shopping Assistant **Aug – Sept 2026**

Client-Side Application — React, TypeScript, Web Speech API, NLP, Vitest

- Built a voice-first shopping assistant using browser speech recognition and deterministic command parsing for English, Hindi, and Hinglish, mapping natural-language requests to **7 shopping intents**.
- Separated NLP/entity extraction from shopping-list, multi-constraint product search, and recommendation modules supporting quantity updates, brand/price filters, purchase-history signals, seasonality, and product substitutes.
- Persisted shopping and activity state with versioned `localStorage` and fallback handling; the documented test suite contains **46 tests** across NLP, shopping, search, recommendations, and persistence.

Technical Skills

Languages	Java, Python, SQL
Backend	Spring Boot, Node.js, Express, REST APIs, JWT
Web Development	React.js, TypeScript
Databases	MongoDB, MySQL, Supabase
Testing / Tools	JUnit 5, MockMvc, Vitest, Maven, Git, Google Colab

Certifications & Achievements

- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — Oracle
- **Oracle Agentic AI Foundations Associate (1Z0-1157-26)** — Oracle
- **Applied Machine Learning in Python** — Certificate
- **TCS CodeVita Season 13 – Global Rank: 1206** — Proof
- **Graph Camp – AlgoUniversity** — Certificate